

FactorJEPA on DENSEWORLD-115k: Can a World Model Predict Crowded, Chaotic South Asian Streets?

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Abstract

Joint Embedding Predictive Architectures (JEPA) forecast the future in embedding space, yet they are mostly evaluated on lower-density, lane-structured traffic. We study DENSEWORLD, the regime of crowded South Asian streets with soft spatial boundaries, heterogeneous agents, persistent occlusion, and negotiated motion, through DENSEWORLD-115k: 115,687 clips (about 1,000 hours) from 22 Indian cities. FactorJEPA replaces the monolithic V-JEPA predictor with separate layout, agent, and interaction channels. After full 115k-clip training it separates from the strongest competing adaptation method on all four diagnostics of predictive structure by 13.9 to 43.3 widths of the paired 95% confidence interval, with rankings that replicate across 2B and 1B backbones (Spearman $\rho \geq 0.895$), while competitors stay ahead on temporal coherence.

Paper — <https://arxiv.org/abs/2608.01049>

Code — <https://github.com/kapilw25/factorjepa>

Dataset — <https://huggingface.co/datasets/anonymousML123/denseworld-115k>

Problem

Video world models such as V-JEPA 2.1 (Mur-Labadia et al. 2026) predict masked or future content as embeddings instead of pixels. Their predictors are monolithic, so nothing forces layout, agents, and interactions to stay distinguishable inside the predicted future, which matters most where these factors are tightly coupled. DENSEWORLD is such a regime: road edges are negotiated rather than marked, pedestrians, two-wheelers, auto-rickshaws, carts, and animals share one right of way, clutter keeps agents partially hidden, and motion follows social negotiation more than lane rules. DENSEWORLD holds 115,687 privacy-filtered clips of 6 to 13 seconds from drive-through, walk-through, and aerial footage (Figure 1, top), and shows consistently higher agent density and occupancy than BDD100K and nuScenes on matched scene types.

*All authors conducted this work independently, outside their roles and employment at their respective companies.

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Key Results

Protocol. We adapt V-JEPA 2.1 at 2B and 1B parameters on a matched stratified 10k-clip set, and the 1B model also on the full 115k-clip corpus. FactorJEPA is compared with the frozen backbone, full fine-tuning, linear probing then fine-tuning (LP-FT), continual self-supervised learning, LoRA (Hu et al. 2022), DoRA (Liu et al. 2024), and Auto-RGN (Lee et al. 2023) under identical clips, masking, steps, and splits. Headline evaluation uses a city-disjoint audit split with independently annotated agents, visibility, interactions, and intervention regions, so no evaluator reuses the training targets.

Diagnostics. Future-frame L1: error between predicted and target future embeddings. Causal L1: whether a controlled edit to an annotated region changes the predicted future as it changes the target. Mask-ratio slope: error growth as evidence is removed. Motion cosine: linear decodability of motion. Margins are in widths of the paired 95% BCa bootstrap confidence interval; 1 or more is separated.

Findings. With limited data (Table 1), FactorJEPA separates on Future-frame L1 and Causal L1 at both scales and on Mask-ratio slope at 1B, while Motion cosine favors generic fine-tuning. Full 115k-clip training reverses that deficit: FactorJEPA then separates on all four diagnostics, so the prediction versus motion trade-off depends on data, not on the factorization. Rankings replicate from the 2B to the roughly half-cost 1B backbone (Spearman $\rho \geq 0.895$). The gain is targeted, not universal: below the rule in Table 1, the best rival stays ahead at 115k on six diagnostics (rollout drift, exposure bias, L1-versus- Δt decay, arrow-of-time, and the two TCC measures) by 8.3 to 146.5 interval widths, so temporal coherence remains the open weakness.

Method

FactorJEPA keeps the pretrained V-JEPA 2.1 encoder, its momentum target encoder, the masking policy, and the prediction targets, and replaces only the predictor. For each target token it predicts layout, agent, and interaction coordinates C_L , C_A , C_I (Figure 1, bottom) and composes the future embedding through learned dictionaries, $\hat{Y} = C_L A_L^\top + C_A A_A^\top + C_I A_I^\top$. A soft visibility gate down-weights occluded agents instead of discarding them, and interaction

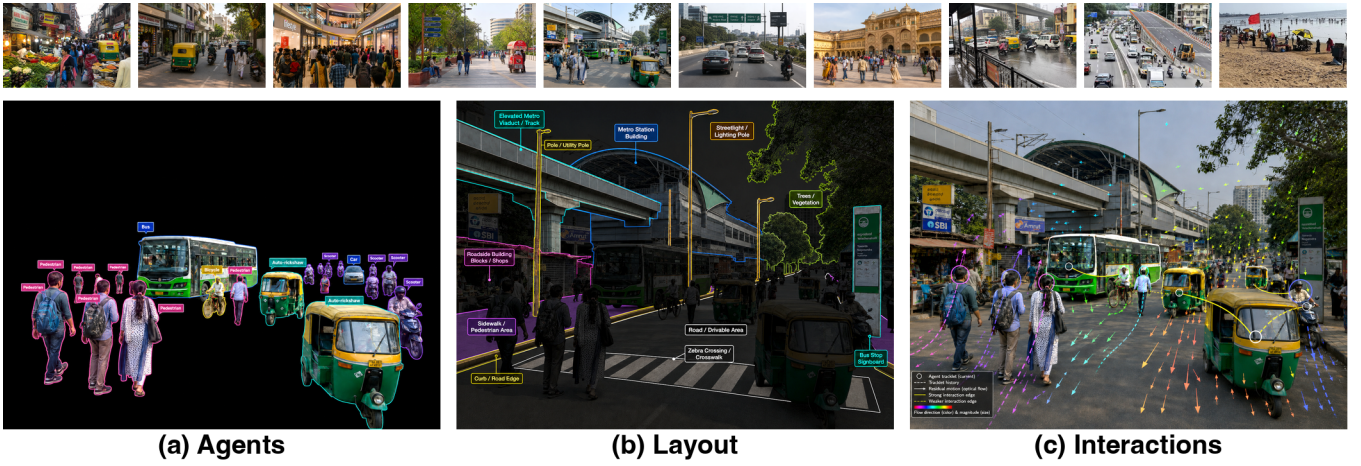


Figure 1: DENSEWORLD at a glance. Top: ten scene types, left to right: market, residential, commercial, promenade, transit, highway, heritage, junction, flyover, and beach, from 115,687 clips across 22 Indian cities. Bottom: one scene factorized into (a) agents, (b) layout with agents suppressed, and (c) interactions: circles mark tracked agents, arrows encode motion (colour: direction; length: magnitude), and solid or dashed edges mark stronger or weaker coupling.

Diagnostic	2B·10k		1B·10k		1B·115k	
	Ours	Rival	Ours	Rival	Ours	Rival
future-frame L1 ($\downarrow$)	0.496	0.504 ^L	0.587	0.596 ^L	0.573	0.589 ^L
causal-block L1 ($\downarrow$)	0.528	0.533 ^L	0.611	0.617 ^L	0.597	0.607 ^L
mask-ratio slope ($\downarrow$)	0.053	0.054 ^L	0.052	0.053 ^D	0.047	0.058 ^L
motion-cos ($\uparrow$)	0.135	0.244^F	0.060	0.090^F	0.097	0.078 ^L
temporal-order ($\uparrow$)	0.953	0.947 ^Z	0.925	0.927^C	0.918	0.915 ^Z
pace ($\uparrow$)	0.736	0.724 ^L	0.731	0.729 ^Z	0.703	0.703 ^L
action top-1 ($\uparrow$)	0.508	0.528^C	0.483	0.509^F	0.567	0.572^L
arrow-of-time ($\uparrow$)	0.964	0.965^Z	0.964	0.965^Z	0.955	0.975^Z
TCC τ ($\uparrow$)	0.288	0.292^Z	0.323	0.334^Z	0.380	0.400^Z
exposure-bias ($\downarrow$)	0.038	0.038 ^C	0.034	0.036 ^D	0.038	0.035^L
rollout drift ($\downarrow$)	0.008	0.008 ^C	0.005	0.005 ^D	0.007	0.006^L
L1-vs- Δt ($\downarrow$)	0.006	0.006 ^C	0.008	0.007^Z	0.009	0.007^Z
TCC cycle ($\downarrow$)	2.092	2.111 ^Z	2.240	2.195^Z	2.806	2.132^Z

Rival superscript: L LoRA, D DoRA, F full fine-tuning, C continual self-supervised learning, Z frozen backbone. At 115k only LoRA and the frozen backbone were trained as rivals.

Table 1: Scores of the best FactorJEPA arm (Ours) and of its strongest rival on all 13 diagnostics, for V-JEPA 2.1 at 2B and 1B on the stratified 10k set and at 1B on the full 115k set. Above the rule: the four diagnostics of predictive structure; below: the rest, where the rival often leads. Bold: the better score of each pair (neither when equal at this precision); arrows give the better direction.

coordinates are sparse weighted sums over candidate agent pairs. Factor heads are anchored to layout, agent, visibility, and interaction targets from a frozen DINOv2 pipeline (Oquab et al. 2024), on the training split only, and a penalty on cross-channel covariance and kernel alignment limits short-cut mixing. The objective adds separation, sparsity, visibility, and factor-supervision terms to the JEPA loss, and only the

predictor and the top K encoder blocks are trained. The work bridges video self-supervised learning, multi-agent scene understanding, and urban mobility in a setting current world-model benchmarks rarely cover.

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